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Survey Paper

Identification of Processes in Closed Loop —Identifiability and Accuracy Aspects*

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An overview of identification of processes operating in closed loop indicates that in many situations such an approach is feasible both from the identifiability and accuracy point of view.

Key Word Index—Closed loop systems; identification; indentifiability; accuracy of identification; optimal experiment design.

Summary—It is often necessary in practice to perform identification experiments on systems operating in closed loop. There has been some confusion about the possibilities of successful identification in such cases, evidently due to the fact that certain common methods then fail. A rapidly increasing literature on the problem is briefly surveyed in this paper, and an overview of a particular approach is given. It is shown that prediction error identification methods, applied in a direct fashion will given correct estimates in a number of feedback cases. Furthermore, the accuracy is not necessarily worse in the presence of feedback; in fact optimal inputs may very well require feedback terms. Some practical applications are also described.

1. *Introduction*

THE PURPOSE of an identification experiment is to determine the dynamics of a given process. Most processes operate as a part of a control configuration and the inputs to the process are partly determined as feedback from other signals. In many cases, security or production reasons do not permit regulators to be removed during an identification experiment. In other cases, such as economic and biological systems, the feedback effects may be inherent. Consequently, identification experiments frequently have to be performed on processes operating in closed loop.

During the last few years there has been a rapidly increasing literature on identification of closed loop systems; and at the 4th IFAC symposium on identification, Tbilisi, 1976, a special session was devoted to these problems. The reason is that the task contains some fallacies, and it is not immediately clear when identification can be successfully performed based on closed loop experiments and how it should be done.

The purpose of this paper is to describe the problems which may arise for closed loop identification and to give an overview over the literature in the field. The paper is based on two IFAC congress/symposium contributions: one paper at the 6th IFAC congress, Boston, 1975, reference [6], mainly treating accuracy aspects for closed loop identification, and one survey over the field for the 4th IFAC symposium on identification, Tbilisi, 1976.

Several different approaches have been taken to the problem of closed loop identification. Various estimation schemes have been suggested, different feedback configurations have been considered and the systems and models have been parameterized in varying ways. Moreover, sometimes the regulator may be considered as part of what is to be identified, and then the input and output are considered as a joint process. This approach, which has been pursued in a series of interesting papers by Caines and Chan [1], [2], and others, is quite natural, e.g. for economic systems, where the feedback mechanism is not open for manipulation. For other processes, in particular for industrial ones, it is natural to assume that the regulator is known, and that it can be chosen freely as long as it yields acceptable performance of the output process.

Consequently the picture of approaches, methods and results is quite rich, and it is of course impossible to give a comprehensive treatment of all aspects in a paper like this. In order to make the paper readable, we have chosen to organize it as follows. In each section we first describe problems, definitions and results which are related to our own activities in the field [3]–[7]. At the same time we also cover a substantial part of the major results in the literature on closed loop identification. Sections 2, 6 and 7 also contain a second part, where we describe and discuss aspects from other papers. In that way the paper can be read as a survey of the field, as well as an overview of one of the approaches taken.

In the next section we shall describe the basic problems connected with closed loop identification. Section 3 contains a discussion of identifiability and accuracy. In Section 4 the influence of the chosen identification method is discussed, while Section 5 deals with the effect of different model structures. Section 6 contains the results on identifiability and accuracy aspects under varying feedback configurations, and there also the major survey of other papers is given. Section 7 reviews some applications of closed loop identification.

2. *Basic concepts*

2.1 *Preliminaries.* The result of an identification experiment depends clearly on several items:

the system,
the model structure or the model parameterization,
the identification method used,
the experimental conditions.

System. In this paper a linear, multivariable, discrete-time, stochastic *system*, $\mathcal{S}$, given the general form

$$y(t) = G_{\mathcal{S}}(q^{-1})u(t) + H_{\mathcal{S}}(q^{-1})e(t) \qquad (2.1)$$

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will be considered. The output, $y(t)$, is a vector of dimension n_y and the input, $u(t)$, has dimension n_u. The variables $\{e(t)\}$ are a sequence of independent, random variables with zero mean values and covariances $Ee(t)e^T(t) = \Lambda$. In (2.1) q^{-1} denotes the backward shift operator, $q^{-1}u(t) = u(t-1)$. It is assumed that $G_{\mathscr{S}}(z)$ and $H_{\mathscr{S}}(z)$ are matrices of appropriate dimensions with rational functions as entries. It is no loss of generality to assume that $e(t)$ has the same dimension as $y(t)$ and that $H_{\mathscr{S}}(0) = I$. Also let $\det [H_{\mathscr{S}}(z)]$ have all zeroes strictly outside the unit circle.

Model. To determine a model of the system the functions $G(z)$ and $H(z)$ have to be parameterized in a suitable manner by a parameter vector θ. A model corresponding to a certain value of θ is denoted by $\mathscr{M}(\theta)$ and is given by

$$y(t) = G_\theta(q^{-1})u(t) + H_\theta(q^{-1})\epsilon(t) \tag{2.2}$$

where $\{\epsilon(t)\}$ is a sequence of independent, random vectors with zero mean values and covariances Λ. When θ is varied over the region of feasible values, equation (2.2) represents a family of models denoted by $\mathscr{M}$. This family will sometimes be referred to as the "*model structure*". The identification problem is to determine the parameter θ so that $\mathscr{M}(\theta)$ in some sense suitably describes the system, $\mathscr{S}$, given by equation (2.1).

The parameterization of $G_\theta(q^{-1})$ and $H_\theta(q^{-1})$ can be made in several ways. Frequently used representations are the' vector difference equation,

$$A_\theta(q^{-1})y(t) = B_\theta(q^{-1})u(t) + C_\theta(q^{-1})\epsilon(t) \tag{2.3}$$

and the state space form, here given in the time-invariant innovations representation form,

$$\begin{cases} x(t+1) = A_\theta x(t) + B_\theta u(t) + K_\theta \epsilon(t) \\ y(t) = C_\theta x(t) + D_\theta u(t) + \epsilon(t) \end{cases} \tag{2.4}$$

Experimental conditions. The input $u(t)$ to the system given by (2.1) can be determined in several ways. It can, as in open loop experiments, be chosen freely by the experiment designer. It can also be determined partly from output feedback by a regulator of a given structure etc. On the whole, the manner in which the input is determined will in this paper be referred to as the *experimental condition*, denoted by $\mathscr{X}$. In this concept other conditions such as the sampling rate, the experiment length etc can be included. These will, however, not be considered here.

In this paper we shall discuss identifiability and accuracy aspects of systems operating in closed loop. A typical such system is shown in Fig. 1.

Several different configurations will be discussed in Section 4. They are essentially covered by the one given in Fig. 1 and correspond to various assumptions about v_i, $i = 1, 2, 3$, and about the character of the regulator. We shall throughout assume that the signals u and y are measurable. We also assume that there is a time delay either in the system or in the regulator. For technological systems, implementation of the feedback loop requires such an assumption. However, for economic and biological systems with 'inherent' feedback it may in certain cases be reasonable not to introduce this assumption. One reason is that the dynamics of these systems may be appreciably faster than the usual sampling rate.

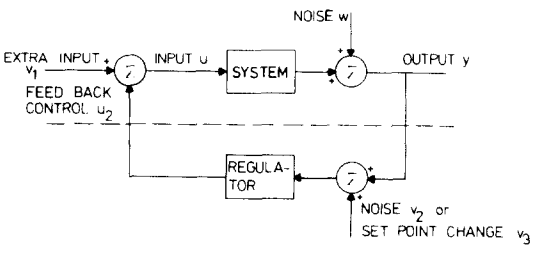


FIG. 1. Block diagram of a typical feedback system.

Identification method. Finally, the procedure by which θ is determined is called the *identification method*, $\mathscr{I}$. For systems operating in closed loop different ways to apply the identification methods are possible. One way which can always be applied is to treat the input–output data exactly as if they were obtained from an open loop experiment. This procedure will be called *direct identification*, $\mathscr{I}_1$.

If for example the regulator is linear, noise-free and time-invariant, or alternates between such feedback laws, then an indirect way of estimating the parameters can be applied. The closed loop system can be regarded as a whole and its parameters can be determined by some method. The open loop process parameters may then be determined from the estimated closed loop process parameters using the knowledge of the regulator. This approach will be called *indirect identification*, $\mathscr{I}_2$.

Further, in some cases it may be advantageous to consider both the input and output processes jointly as the output of a system driven by noise only. This approach may be called *joint input–output identification*, $\mathscr{I}_3$.

Within each of these approaches, different methods can be used, such as correlation and spectral analysis, various parametric identification methods, etc. We shall in particular consider prediction error identification methods, cf. [8]. For these methods the parameter vector θ in the model set (2.2) is chosen so that the scalar function

$$g(Q_N(\mathscr{S}, \mathscr{M}(\theta))) \tag{2.5a}$$

of the matrix

$$Q_N(\mathscr{S}, \mathscr{M}(\theta)) = \frac{1}{N}\sum_1^N [y(t) - \hat{y}(t|t-1; \mathscr{M}(\theta))] \times [y(t) - \hat{y}(t|t-1; \mathscr{M}(\theta))]^T \tag{2.5b}$$

is minimized. Here $\hat{y}(t|t-1; \mathscr{M}(\theta))$ denotes the linear least squares prediction of $y(t)$ based on data up to time $t-1$, and using the model $\mathscr{M}(\theta)$. Neglecting initial value effects it is given by

$$\hat{y}(t|t-1; \mathscr{M}(\theta)) = [I - H_n^{-1}(q^{-1})]y(t) + H_n^{-1}(q^{-1})G_n(q^{-1})u(t). \tag{2.6}$$

The 'true' prediction $\hat{y}(t|t-1; \mathscr{S})$ is defined analogously. The function g in (2.5a) maps symmetric $n_y \times n_y$ matrices into the real line. Common choices of g are 'trace' or 'determinant'. If g is chosen as 'determinant' which in a certain sense is the optimal choice, see [84], then, for the below consistency result it is required that the covariance matrix of the innovations, $y(t) - \hat{y}(t|t-1; \mathscr{S})$ is non-singular.

Let the parameter vector θ that minimizes (2.5), the estimate at time N, be denoted by $\theta(N; \mathscr{S}, \mathscr{M}, \mathscr{I}, \mathscr{X})$. In Ljung[8] it is shown that under mild conditions

$$\theta(N; \mathscr{S}, \mathscr{M}, \mathscr{I}, \mathscr{X}) \to D_I(\mathscr{S}, \mathscr{M}, \mathscr{I}, \mathscr{X}) \text{ with prob one as } N \to \infty$$

where $D_I(\mathscr{S}, \mathscr{M}, \mathscr{I}, \mathscr{X})$ is the set of such θ, that yield a model-predictor which asymptotically behaves like the true predictor in the sense that

$$D_I(\mathscr{S}, \mathscr{M}, \mathscr{I}, \mathscr{X}) = \{\theta | \lim_{N\to\infty} \frac{1}{N}\Sigma E|\hat{y}(t|t-1; \mathscr{S}) - \hat{y}(t|t-1; \mathscr{M}(\theta))|^2 = 0\}. \tag{2.7}$$

It is also clear that 'desired estimates' are given by the set

$$D_T(\mathscr{S}, \mathscr{M}) = \{\theta | G_\theta(z) = G_{\mathscr{S}}(z) \text{ and } H_\theta(z) = H_{\mathscr{S}}(z) \text{ a.e. } z\}. \tag{2.8}$$

This set consists of the parameter values that give models $\mathscr{M}(\theta)$ with the same transfer function and the same noise characteristics as the system $\mathscr{S}$.

It should be noted that in case $C_\theta = I$ in (2.3), then (2.5) is nothing but the common least squares estimation method. Further, for general C-polynomials, but under the reasonable assumption that $e(\cdot)$ in (2.1) and $\epsilon(\cdot)$ in (2.2) are gaussian

processes, then (2.5) with $g(\cdot) = \det(\cdot)$ defines the maximum likelihood estimate of the system parameters, cf. [8]. This is a very widely used identification method, introduced into dynamic system identification in [23].

2.2 *Basic problems.* There are several interesting problems concerning identification of systems operating in closed loop. Some of them will be reviewed here.

Identifiability. The question of identifiability concerns whether or not the open loop characteristics of the system can be obtained as the number of data tends to infinity. The problems when identifying systems operating under closed loop can be illustrated by the following examples.

Example 2.1. Consider the system given in Fig. 2. Suppose the identification is performed by methods which do not imply a causal structure, e.g. methods based on correlation analysis and spectral analysis. Then an attempt to identify $G_{\mathcal{S}}$ from measurements of u and y in a direct fashion will give

$$\hat{G}_{\mathcal{S}} = \frac{1}{G_{\mathcal{R}}}$$

i.e. the inverse of the transfer function of the feedback. This has been pointed out e.g. by Akaike[9], Åström–Eykhoff[10], and Box–MacGregor[11]. □

Using other identification methods, e.g. a prediction error identification method and postulating a causal parametric model, identifiability may be achieved for this type of experimental condition. This is not always true which can be seen from the next example.

Example 2.2. Consider the system

$$y(t) + ay(t-1) = bu(t-1) + e(t) \tag{2.9}$$

with a proportional regulator

$$u(t) = gy(t).$$

An attempt to estimate the parameters a and b, e.g. by the least squares method, shows that all parameter estimates

$$\begin{cases} \hat{a} = a + \gamma g \\ \hat{b} = b + \gamma \end{cases}$$

γ arbitrary, give the same value of the identification criterion. For $\gamma \neq 0$ an erroneous description of the open loop system is obtained. Notice in particular that it is of no help to know the regulator parameter g.

If on the other hand identification is considered in a class of models given by

$$y(t) + ay(t-1) = \hat{b}u(t-1) + \epsilon(t)$$

where a is known, it is obvious that the system (2.9) will be identifiable. □

The examples have shown the following characteristic problems when identifying systems operating in closed loop,

—the importance to choose an appropriate identification method (see Example 2.1),

—the importance of using experimental conditions which will guarantee identifiability (in Example 2.2 e.g. shifting between two proportional regulators with different g will do),

—the importance of choosing a proper model structure in cases where the experimental conditions do not guarantee identifiability (see Example 2.2).

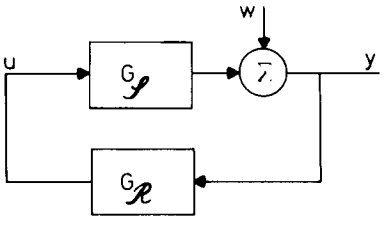


FIG. 2. A simple closed loop configuration.

Testing of identifiability. As will be seen in Section 4 some conditions invoking the true order of the system must be evaluated in order to test identifiability for certain kinds of experimental conditions. Thus *a priori* information of the order of the system is a crucial point. Moreover, the order cannot be tested *a posteriori*, i.e. it is not possible to test after the experiment if the system is actually identifiable. This fact has been pointed out e.g. by Bohlin[12] and can be illustrated by the following example.

Example 2.3. Consider the system

$$A(q^{-1})y(t) = q^{-k}B(q^{-1})u(t) + C(q^{-1})e(t) \tag{2.10}$$

with the feedback given by

$$F(q^{-1})u(t) = G(q^{-1})y(t). \tag{2.11}$$

It is obvious that the system

$$[A(q^{-1}) + L(q^{-1})G(q^{-1})]y(t) = \\ = [q^{-k}B(q^{-1}) + L(q^{-1})F(q^{-1})]u(t) + C(q^{-1})e(t) \tag{2.12}$$

with the feedback (2.11) and with an arbitrary polynomial $L(q^{-1})$ will have the same input–output relation in closed loop as the system (2.10). Hence from input–output experiments (under the given experimental conditions: a linear, noise-free, time-invariant regulator) the true system (2.10) cannot be distinguished from (2.12). This means that it cannot in general be established *a posteriori* that the obtained system is identifiable and hence that the estimated model has the same open loop transfer function as the system. □

Accuracy aspects. So far only identifiability has been considered. In practice also the quality of the estimates, i.e. the accuracy, for a finite number of data is of interest. It is then an important question how the experimental condition influences the accuracy. This problem is closely related to optimal experiment design and can be viewed as an extension of the optimal input synthesis problem which has been given much attention. Regarding this problem see e.g. the survey paper, Mehra[13].

2.3. *Other formulations and problems.* The preliminaries given in Section 2.1 cover most of the approaches in the literature. The system and models are almost always taken to be linear. Usually more specific linear models than (2.2) are considered, e.g. (2.3), (2.4) ([14]–[16]), impulse response models ([17]–[19]) or transfer function models ([19], [19]–[22]), cf. Table 6.1 below. Most experiment configurations studied are covered by Fig. 1. There may be some more specific ones discussed, and it should be realized that identifiability properties may be critically dependent upon what is known about the special structure of the system vs the regulator. Some schemes, like indirect identification assume that v_1, rather than u is measureable. The most diverse situation is for the choice of estimation scheme, $\mathcal{J}$. Apart from prediction error identification methods, (2.5), of which least squares is a special case, correlation and spectral analysis methods are often discussed. They cannot be used in a direct identification fashion, cf. Example 2.1, but several special schemes have been deviced and discussed, e.g. Akaike[9], [17], Box and MacGregor[11], Rödder[20].

Problems, in addition to those listed in Section 2.2, which have been discussed are basically the problem to detect whether feedback is present and to analyse causal relationships between the recorded signals. These questions are treated, e.g. in [2].

3. *Identifiability and measures of accuracy*

3.1. *Identifiability.* The concept of identifiability has been given several different definitions in the literature. The most common approach is to relate the identifiability property to consistency of the parameter estimate $\hat{\theta}(N)$. The 'true' parameter θ_0 is then said to be identifiable, if the sequence of estimates $\hat{\theta}(N)$ converges to θ_0 in some stochastic sense, see, e.g., Åström-Bohlin[23], Staley-Yue[24] and Tse-Anton[25]. In this paper the definition introduced in Ljung *et al.*[4] will be used (D_T is the set defined by (2.8)).

Definition 1. The system $\mathcal{S}$ is said to be *System Identifiable under* $\mathcal{M}$, $\mathcal{I}$ *and* $\mathcal{X}$, $SI(\mathcal{M}, \mathcal{I}, \mathcal{X})$, if $\theta(N; \mathcal{S}, \mathcal{M}, \mathcal{I}, \mathcal{X}) \to D_T(\mathcal{S}, \mathcal{M})$ with prob one as $N \to \infty$, [i.e. $\inf_{a \in D_T} |\theta(N; \mathcal{S}, \mathcal{M}, \mathcal{I}, \mathcal{X}) - \theta| \to 0$ w.p. 1 as $N \to \infty$].

Definition 2. The system $\mathcal{S}$ is said to be *Strongly System Identifiable under* $\mathcal{I}$ *and* $\mathcal{X}$, $SSI(\mathcal{I}, \mathcal{X})$, if it is $SI(\mathcal{M}, \mathcal{I}, \mathcal{X})$ for all $\mathcal{M}$ such that $D_T(\mathcal{S}, \mathcal{M})$ is non-empty.

Definition 3. The system $\mathcal{S}$ is said to be *Parameter Identifiable under* $\mathcal{M}$, $\mathcal{I}$ *and* $\mathcal{X}$, $PI(\mathcal{M}, \mathcal{I}, \mathcal{X})$, if it is $SI(\mathcal{M}, \mathcal{I}, \mathcal{X})$ and $D_T(\mathcal{S}, \mathcal{M})$ consists of only one element.

The concepts can be illustrated by the following example.

Example 3.1.(*cf. Example* 2.1 *etc*)

Consider the system, $\mathcal{S}$,

$$y(t) = bu(t-1) + e(t). \tag{3.1}$$

First let the experimental condition be that the system is governed by the feedback law, $\mathcal{X}_1$,

$$u(t) = gy(t). \tag{3.2}$$

Assume that the model structure, $\mathcal{M}_1$,

$$y(t) = \hat{b}u(t-1) + \epsilon(t)$$

is used and that the parameter $\hat{b}$ is estimated using the least squares method, $\mathcal{I}_{LS}$. It is easy to see that $\hat{b}$ will converge to b. Hence the system $\mathcal{S}$ is $SI(\mathcal{M}_1, \mathcal{I}_{LS}, \mathcal{X}_1)$ as well as $PI(\mathcal{M}_1, \mathcal{I}_L S_S, \mathcal{X}_1)$.

Assume then that the model structure, $\mathcal{M}_2$,

$$y(t) + \hat{a}y(t-1) = \hat{b}u(t-1) + \epsilon(t) \tag{3.3}$$

is used. Inserting the feedback law (3.2) in (3.3) gives

$$y(t) + (\hat{a} - g\hat{b})y(t-1) = \epsilon(t).$$

Hence all $\hat{a}$ and $\hat{b}$ such that

$$\hat{a} - g\hat{b} = -gb$$

will give a correct description of the closed loop system (3.1) with the feedback law (3.2). The set $D_t(\mathcal{S}, \mathcal{M}_2, \mathcal{I}_{LS}, \mathcal{X}_1)$ therefore consists of all $\hat{a}$ and $\hat{b}$ satisfying this equation. However, obviously e.g.

$$\hat{a} = 1$$
$$\hat{b} = b + 1/g$$

in (3.3) does not give the correct transfer function of the system $\mathcal{S}$. The system $\mathcal{S}$ is thus neither $SI(\mathcal{M}_2, \mathcal{I}_{LS}, \mathcal{X}_1)$ nor $PI(\mathcal{M}_2, \mathcal{I}_{LS}, \mathcal{X}_1)$. In particular $\mathcal{S}$ is not $SSI(\mathcal{I}_{LS}, \mathcal{X}_1)$ for the experimental configuration, i.e. the feedback law, under consideration.

Now let the experimental condition be that the input is white noise, uncorrelated with $\{e(t)\}$, i.e. the system is operating in open loop. Let this experimental condition be denoted by $\mathcal{X}_2$. Let the model structure $\mathcal{M}_3$ be given by the set of models

$$y(t) + \hat{a}y(t-1) = \hat{b}_1 u(t-1) + \hat{b}_2 u(t-2) + \epsilon(t) + \hat{c}\epsilon(t-1). \tag{3.4}$$

The maximum likelihood method, $\mathcal{I}_{ML}$, is used to estimate the parameters $\hat{a}$, $\hat{b}_1$, $\hat{b}_2$ and $\hat{c}$. It can be shown that this method gives estimates that converge to values satisfying

$$\hat{a} = \mu$$
$$\hat{b}_1 = b$$
$$\hat{b}_2 = \mu b$$
$$\hat{c} = \mu$$

where μ is arbitrary. $\mathcal{S}$ is thus not $PI(\mathcal{M}_3, \mathcal{I}_{ML}, \mathcal{X}_2)$. However, the transfer function of (3.4) for these parameter values is the correct one for any value of μ. Consequently, $\mathcal{S}$ is $SI(\mathcal{M}_3, \mathcal{I}_{ML}, \mathcal{X}_2)$. In fact, for this experimental configuration, $\mathcal{X}_2$, this choice of input signal, the system $\mathcal{S}$ is $SSI(\mathcal{I}_{ML}, \mathcal{X}_2)$. □

The definitions above make no reference to any 'true' parameter value, θ_0, but should be regarded as consistency-oriented since the requirement that $D_T(\mathcal{S}, \mathcal{M})$ is non-empty implies that there is a 'very good' model available among the set of models $\mathcal{M}$. If the set of permitted values θ contains a 'true' parameter θ_0 then the definition of parameter identifiability is equivalent to the definitions given by e.g. Tse–Anton[25], Glover–Willems[26] and Bellman–Åström[27].

Notice that $PI(\mathcal{M}, \mathcal{I}, \mathcal{X})$ is always implied by $SI(\mathcal{M}, \mathcal{I}, \mathcal{X})$ if $D_T(\mathcal{S}, \mathcal{M})$ consists of only one point. This condition on $D_T(\mathcal{S}, \mathcal{M})$ involves neither $\mathcal{I}$ nor $\mathcal{X}$, and is the problem of canonical representation of transfer functions. It turns out to be convenient to treat this problem separately and study the identifiability properties for different experimental conditions, identification methods and model structures by considering $SI(\mathcal{M}, \mathcal{I}, \mathcal{X})$.

Clearly, a necessary condition on $\mathcal{M}$ to achieve $SI(\mathcal{M}, \mathcal{I}, \mathcal{X})$ is that $D_T(\mathcal{S}, \mathcal{M})$ is non-empty. If the system is $SSI(\mathcal{I}, \mathcal{X})$, this condition is also a sufficient condition on $\mathcal{M}$ for $SI(\mathcal{M}, \mathcal{I}, \mathcal{X})$. In that case the fact that the system may operate in closed loop does not add any extra difficulties when choosing appropriate model structures $\mathcal{M}$. Experimental conditions that give $SSI(\mathcal{I}, \mathcal{X})$ therefore are equivalent to open loop from the view point of identifiability.

3.2. *Criteria of accuracy.* The reasons for performing parameter identification may for example be: determination of certain parameter values which have some specific physical meaning, control system design, or prediction.

The result of an identification is very often used for all the above purposes. This immeidately leads to the conclusion that it is not possible to give a single criterion of accuracy covering all situations. Discussions about different criteria and references to other works on this subject can be found in Mehra[13], and e.g. Söderström *et al.*[6].

Let us assume that the system is $PI(\mathcal{M}, \mathcal{I}, \mathcal{X})$ under the chosen conditions, so that $\theta(N; \mathcal{S}, \mathcal{M}, \mathcal{I}, \mathcal{X})$ converges w.p.1 to the 'true value' θ_0. One common measure of accuracy is then to use the covariance matrix P of the parameter estimates. The asymptotic covariance matrix of the estimates can be computed for various cases, and it depends of course among other things on the identification method used. For prediction error identification methods the asymptotic covariance is determined in [84] for different functions $g(\cdot)$ under mild conditions, e.g. not including any assumption about the distribution of the innovations. The function $g(\cdot) = \det(\cdot)$ gives the best accuracy (and so does $g(A) = tr(A\Lambda^{-1})$ if Λ is known),

$$P = P(N) \sim \frac{1}{N}[E\epsilon'(t)^T \Lambda^{-1} \epsilon'(t)]^{-1} \tag{3.5}$$

where $\epsilon'(t)$ denotes the derivative of $\epsilon(t)$ with respect to θ, see (2.3), and evaluated in θ_0. N is the number of data processed and $\sim$ denotes 'asymptotically equal to'. In fact, P is to be regarded as the variance of the limiting (normal) distribution of $\theta(N; \mathcal{S}, \mathcal{M}, \mathcal{I}, \mathcal{X}) - \theta_0$ rather than as the variance of $\theta(N; \mathcal{S}, \mathcal{M}, \mathcal{I}, \mathcal{X})$. In the sequel we shall on several occasions evaluate the asymptotic variance P for different cases, in this case expression (3.5) will always be used.

If the distribution of the innovations is normal then the prediction error method with $g(\cdot) = \det(\cdot)$ gives the maximum likelihood estimate and the covariance matrix P given by (3.5) is then also the Cramer–Rao lower bound for the variance of *any* unbiased estimate of θ_0, cf. [29], [30], and [84]. Its inverse is then the *information matrix*, see e.g. Kendall–Stuart[28] for a definition, and it will be denoted by J,

$$J = P^{-1} \quad \text{(for gaussian innovations)}. \tag{3.6}$$

The inverse of the information matrix is always the Cramer–Rao lower bound, and it can therefore be used as a

measure of accuracy that is independent of the identification method. It is however difficult to compute J for general innovations, and we shall in the sequel work with P, computed according to (3.5).

It is however advantageous to have a *scalar* measure. Two different scalar measures based on P will be used in this paper. One is $\det P$, which is commonly used, see e.g. Mehra[13] and Nahi and Napjus[31]. The smaller the value of $\det P$ is, the better the identification result. The other measure was described in Söderström[85]. The idea is to use a scalar, twice differentiable function $V(\theta)$ such that

$$V(\theta_0) = \inf V(\theta) \tag{3.7}$$

where, as before, θ_0 is the true value. Note that it is not necessary that $V(\cdot)$ has a unique global minimum point. Two examples of such functions will now be given for single output systems.

Example 3.2. Let

$$V_a(\theta) = E\epsilon^2(t). \tag{3.8}$$

The loss function $V_a(\theta)$ can be interpreted as the variance of the one step ahead prediction error that will be obtained using the model $\mathcal{M}(\theta)$. □

Example 3.3. Assume that the model $\mathcal{M}(\theta)$ is used for construction of a minimum variance regulator, see Åström[36]. Suppose that the true system is controlled by this regulator and take

$$V_b(\theta) = Ey^2(t). \tag{3.9}$$ □

Note that $V(\theta)$ expresses the obtained accuracy for one specific model (obtained from one specific identification experiment). To obtain a realization-free measure, expectation will be taken as follows.

Close to the true parameter θ_0, where V has a minimum, one gets

$$V(\theta) \cong \tilde{V}(\theta) \overset{\Delta}{=} V(\theta_0) + \tfrac{1}{2}(\theta - \theta_0)^T V''(\theta_0)(\theta - \theta_0).$$

As a suitable measure of the accuracy thus the scalar

$$E\tilde{V}(\theta(N; \mathcal{S}, \mathcal{M}, \mathcal{I}, \mathcal{X})) = V(\theta_0) + \tfrac{1}{2} tr(V''(\theta_0)P) \tag{3.10}$$

can be chosen.

In (3.10) $V(\theta_0)$ is a constant independent of $\mathcal{M}$, $\mathcal{I}$ and $\mathcal{X}$. Note the similarity with a criterion of the form tr AP, where A is a weighting matrix, see e.g. Mehra[13]. The proposed technique gives a possibility to generate weighting matrices having simple physical interpretations.

The application of the measure $E\tilde{V}(\theta(N; \mathcal{S}, \mathcal{M}, \mathcal{I}, \mathcal{X}))$ for comparing different $\mathcal{I}$ or $\mathcal{X}$ is straightforward. To compare different $\mathcal{M}$ there is a formal difficulty since then two different parameter vectors of different sizes may be used. Assume that θ_1 and θ_2 are to be compared and that

$$\mathcal{M}_1 \subset \mathcal{M}_2. \tag{3.11}$$

Let the vector θ_i correspond to $\mathcal{M}_i$ and denote the function V for this case by $V_i(\theta_i)$. The relation (3.11) is well defined, since $\mathcal{M}_i$ are sets. A possible interpretation can be that θ_1 is obtained by fixing some parameters in θ_2. We can regard $\mathcal{M}_1$ as a restriction of $\mathcal{M}_2$. Doing so, we can describe the set $\mathcal{M}_1$ by θ_2 parameterized with θ_1, i.e. θ_2 is to be a function $\theta_2 = \theta_2(\theta_1)$.

Example 3.4. Consider the model structures

$$\mathcal{M}_1\colon\ y(t) + a_1 y(t-1) = \epsilon(t) \qquad \theta_1 = a_1$$

$$\mathcal{M}_2\colon\ y(t) + a_1 y(t-1) + a_2 y(t-2) = \epsilon(t) \qquad \theta_2 = [a_1 \quad a_2]^T.$$

Then clearly $\mathcal{M}_1 \subset \mathcal{M}_2$. The function $\theta_2 = \theta_2(\theta_1)$ becomes $\theta_2(\theta_1) = [\theta_1 \quad 0]^T$. □

To do the comparison between $V_1(\theta_1)$ and $V_2(\theta_2)$ it is required that

$$V_1(\theta_1) = V_2(\theta_2(\theta_1)) \tag{3.12}$$

for all relevant θ_1.

4. *Influence of the identification method on identifiability and accuracy*

The influence of the identification method on identifiability and especially on the accuracy of the obtained results is a widely studied problem in the identification literature. Here we shall concentrate on the importance of choosing an identification method for closed loop identification. As mentioned in Section 2.1, three different approaches can be distinguished; direct identification, $\mathcal{I}_1$, indirect identification, $\mathcal{I}_2$ and joint input–output identification, $\mathcal{I}_3$. Within each of these approaches different identification and estimation methods can be used. It should be remarked that while $\mathcal{I}_1$ and $\mathcal{I}_3$ do not impose any restrictions on the experimental condition, indirect identification $\mathcal{I}_2$ requires that the regulator is known and noise-free ($v_2 = 0$). Further, if extra input v_1 is present, it must be measurable. It should also be noted that even through $\mathcal{I}_3$ is applicable in principle to arbitrary feedback configurations, it does require a parametric model of the feedback mechanism, and therefore $\mathcal{I}_3$ may be unfeasible in practice for, e.g. nonlinear feedback. Such restrictions do not apply to $\mathcal{I}_1$.

In the remainder of this section we shall discuss some identifiability and accuracy properties obtained for $\mathcal{I}_1$, $\mathcal{I}_2$ and $\mathcal{I}_3$ in connection with various estimation schemes.

4.1. *Influence on identifiability.* As shown already in Example 2.1, direct identification $\mathcal{I}_1$, used with correlation or spectral analysis will not yield identifiability. Therefore, we shall always assume that $\mathcal{I}_1$ is used with a prediction error identification method, that assumes a causal relationship between the input and the output.

Joint input–output identification, $\mathcal{I}_3$, can be used in a straightforward manner together with a prediction error method, e.g. [1], [2], [58], or with specially designed correlation techniques, e.g. [9]. Let us study in somewhat more detail how $\mathcal{I}_3$ works with a prediction error method. Assume that the model of the input–output process is

$$\hat{y}(t|t-1; \mathcal{M}(\theta)) = f_1(\mathcal{Y}_{t-1}, \mathcal{U}_{t-1}; \mathcal{M}(\theta)) \tag{4.1a}$$

$$\hat{u}(t|t-1; \mathcal{M}(\theta)) = f_2(\mathcal{Y}_{t-1}, \mathcal{U}_{t-1}; \mathcal{M}(\theta)) \tag{4.1b}$$

where $\mathcal{Y}_{t-1} = \{y(t-1), \ldots, y(0)\}$ and $\mathcal{U}_{t-1} = \{u(t-1), \ldots, u(0)\}$. If there is a time delay in the model (4.1a), then this expression is also the prediction model for direct identification. Therefore, for $\mathcal{I}_1$ some function of $y(t) - \hat{y}(t|t-1; \mathcal{M}(\theta))$ is minimized while for $\mathcal{I}_3$ the vector

$$\begin{bmatrix} y(t) - \hat{y}(t|t-1; \mathcal{M}(\theta)) \\ u(t) - \hat{u}(t|t-1; \mathcal{M}(\theta)) \end{bmatrix}$$

is considered. Let us assume that the process model, f_1, and the feedback model f_2 do not have any parameters in common, which is the normal case. Let us also assume that the noise source v_2 (if any) in the regulator is independent of the process noise w, see Fig. 2.1 or, more precisely that $v_2(t)$ is independent of future $w(s)$, $s \geq t$. Then predictions of u never enter into y, due to the time delay, and the model (4.1b) does not at all affect the result in modelling (4.1a). If g in (2.5a) is chosen as 'trace' for $\mathcal{I}_1$ and $\mathcal{I}_3$, we minimize for $\mathcal{I}_1$

$$\frac{1}{N}\sum_1^N |y(t) - \hat{y}(t|t-1; \mathcal{M}(\theta))|^2$$

and for $\mathcal{I}_3$

$$\frac{1}{N}\sum_1^N |y(t) - \hat{y}(t|t-1; \mathcal{M}(\theta))|^2 + \frac{1}{N}\sum_1^N |u(t) - \hat{u}(t|t-1; \mathcal{M}(\theta))|^2.$$

However, as just explained the second term in the last expression does not influence the first one, and therefore the result from $\mathcal{I}_1$ is identical to that of $\mathcal{I}_3$ for (4.1a). If there is no time delay in the system but there is one in the regulator, we may shift $u(t)$ one time step and repeat the discussion. If there is no time delay neither in the regulator nor in the system, then there is a difference in the approaches $\mathcal{I}_1$ and $\mathcal{I}_3$.

For indirect identification, $\mathcal{I}_2$, the dynamics of the closed loop system as a whole is determined first. Since this system

is operating in 'open loop', no restriction regarding the estimation methods have to be introduced; for example, correlation methods will work here. In the second phase of $\mathcal{I}_2$ the open loop dynamics is solved for from the closed loop dynamics and from the knowledge of the regulator. This can be computationally complicated and it may or may not be successful, depending on the experimental conditions. For the quite general case, where the regulator shifts between a finite number of fixed linear regulators and $v_2 = 0$ and v_1 either is zero or measurable, we have the result, Theorem 6.1 below, (see also [33] and [4]) that the identifiability properties of $\mathcal{I}_1$, used with a prediction error method, coincide with those of $\mathcal{I}_2$, used with any identification method.

To summarize this discussion:

—$\mathcal{I}_1$ must be used with an estimation method that assumes a causal system, such as a prediction error method.

—$\mathcal{I}_1$ and $\mathcal{I}_3$ used with a prediction error method give identical system models if there is a time delay either in the regulator or in the system, and the regulator noise is independent of the system noise.

—$\mathcal{I}_1$ with a prediction error method and $\mathcal{I}_2$ with any identification method have identical identifiability properties.

4.2. *Influence on the accuracy.* It follows from the previous subsection that the accuracies for $\mathcal{I}_1$ and $\mathcal{I}_3$ used with prediction error methods are identical, if the process model and the feedback model do not have any parameters in common. Moreover, if the noise sources in the process are gaussian, then $\mathcal{I}_1$ and $\mathcal{I}_3$ give the maximum likelihood estimates of the system parameters if the function $g(\cdot)$ in (2.5a) is chosen as $\det(\cdot)$. Then, as explained in Section 3.2, the estimates are asymptotically efficient, i.e. no other way of obtaining unbiased estimates of the system parameters can give a smaller covariance matrix. Consequently, $\mathcal{I}_1$ is 'the best possible' method in this sense.

A comparison between direct identification, $\mathcal{I}_1$ and indirect, $\mathcal{I}_2$ (using a prediction error method to determine the parameters of the closed loop system) is made in [7]. For a single-input single-output system with the experimental condition making indirect identification applicable, the open loop parameters can in general be determined from the closed loop ones by solving an overdetermined system of linear equations. If this overdetermined system is solved using the Markov estimate, then it can be shown that the covariance matrices obtained by $\mathcal{I}_1$ and $\mathcal{I}_2$ respectively are the same. The calculations are given in detail in [7].

The result means that the considerably more involved procedure of indirect identification has no advantage over direct identification, neither from the viewpoint of identifiability nor accuracy.

5. *Influence of the model structure on identifiability and accuracy*

5.1 *Influence on identifiability.* The choice of model structure, $\mathcal{M}$, can be dictated by various considerations. First of all, the objective of the model plays an important role. A model that shall be used for regulator design is parameterized differently from a model which is determined to increase the physical insight into the process or to find certain physical constants. The basic problem is then to choose an $\mathcal{M}$, that is flexible enough to allow a description of the true system; i.e. to make $D_T(\mathcal{S}, \mathcal{M})$ non-empty, but small enough so that D_T contains just one element, i.e. to assure $PI(\mathcal{M}, \mathcal{I}, \mathcal{X})$. If the objective of the identification is to find a model for simulation or control synthesis, we may sometimes disregard the second problem, and be satisfied if $SI(\mathcal{M}, \mathcal{I}, \mathcal{X})$ is obtained. Now, if the system is Strongly system identifiable, $SSI(\mathcal{I}, \mathcal{X})$, we obtain $SI(\mathcal{M}, \mathcal{I}, \mathcal{X})$ if $D_T(\mathcal{S}, \mathcal{M})$ is non-empty, and the model set can never be too large. Otherwise, however, an increased flexibility of the model set possibly implies that the *SI*-property is lost, cf. Example 3.1. This is further discussed in [5], where exact conditions on the model set are given for a common closed loop identification problem, see also [87], [88].

This short discussion summarizes the effects of the choice of $\mathcal{M}$ upon the identifiability properties. The effects are essentially condensed in our identifiability definitions in Section 3.

5.2 *Influence on the accuracy.* If we are particularly interested in $PI(\mathcal{M}, \mathcal{I}, \mathcal{X})$, we may proceed to investigate how the accuracy of the estimates is influenced by the choice of $\mathcal{M}$. Often, several different $\mathcal{M}$ yield $PI(\mathcal{M}, \mathcal{I}, \mathcal{X})$ and we would then be interested in those that give the best accuracy.

Theorem 5.1. Consider two model structures $\mathcal{M}_1$ and $\mathcal{M}_2$ which *both* make the system parameter identifiable. Assume that

$$\mathcal{M}_1 \subset \mathcal{M}_2 \tag{5.1}$$

and that we have a twice differentiable function $V(\theta)$ satisfying (3.7), (3.12). Then

$$E\tilde{V}(\theta(N; \mathcal{S}, \mathcal{M}_1, \mathcal{I}, \mathcal{X})) \leq E\tilde{V}(\theta(N; \mathcal{S}, \mathcal{M}_2, \mathcal{I}, \mathcal{X})). \tag{5.2}$$

Proof. Due to the end of Section 3.2 we can describe the set $\mathcal{M}_1$ via the function $\theta_2(\theta_1)$. Equation (3.12) now implies

$$\frac{d^2 V_{\mathcal{M}_1}}{d\theta_1^2}(\theta_{10}) = \left[\frac{d\theta_2}{d\theta_1}(\theta_{10})\right]^T \times \frac{d^2 V_{\mathcal{M}_2}}{d\theta_2^2}(\theta_{20}) \left[\frac{d\theta_2}{d\theta_1}(\theta_{10})\right] \tag{5.3}$$

and

$$P_{\mathcal{M}_1}^{-1} = \left[\frac{d\theta_2}{d\theta_1}(\theta_{10})\right]^T P_{\mathcal{M}_2}^{-1} \left[\frac{d\theta_2}{d\theta_1}(\theta_{10})\right] \tag{5.4}$$

where $P_{\mathcal{M}_i}$ denotes the covariance matrix when the model structure $\mathcal{M}_i$ is used. The theorem follows from the matrix lemma

$$\operatorname{tr} AB^{-1} \geq \operatorname{tr} (R^T AR)(R^T BR)^{-1}$$

This lemma is proven in [7]. □

Example 5.1. For the special test function V_a of Example 3.2 the more explicit result

$$E\tilde{V}_a(\theta(N; \mathcal{S}, \mathcal{M}, \mathcal{I}, \mathcal{X})) = \Lambda\left(1 + \frac{1}{N} \dim \theta\right)$$

is obtained. □

The conclusion of the theorem is most natural and in accordance with standard estimation results, see e.g. Box and Jenkins [86]. To get the best accuracy, the model should contain as few parameters as possible. Notice that the result is quite general. It holds for multivariable systems, for any experimental conditions and for general model structures.

6. *Influence of feedback on identifiability and accuracy*

In this section the influence of the experimental conditions, $\mathcal{X}$, will be discussed. In one sense $\mathcal{X}$ will have a greater influence on the results than $\mathcal{M}$ and $\mathcal{I}$. If it is found that an $\mathcal{X}$ does not yield identifiability or acceptable accuracy, then the whole identification experiment has to be repeated. In contrast, if $\mathcal{M}$ or $\mathcal{I}$ was not chosen in a suitable manner it is possible to try other $\mathcal{M}$ or $\mathcal{I}$ on the same data.

6.1 *Influence on identifiability.* It follows from (2.6) and (2.7) that the set D_I consists of those θ for which

$$\lim_{N \to \infty} \frac{1}{N} \sum_1^N |L_\theta(q^{-1})y(t) - M_\theta(q^{-1})u(t)|^2 = 0 \tag{6.1}$$

where

$$L_\theta(z) = H_\theta^{-1}(z) - H_{\mathcal{I}}^{-1}(z) \tag{6.2a}$$

$$M_\theta(z) = H_\theta^{-1}(z)G_\theta(z) - H_{\mathcal{I}}^{-1}(z)G_{\mathcal{I}}(z). \tag{6.2b}$$

Clearly, $D_T \subset D_I$ (D_T defined by (2.8)), since for $\theta \in D_T$ we have $L_\theta(z) = M_\theta(z) \equiv 0$. However, if the system is not *SI* under the current experimental conditions, then there exists a $\theta^* \in D_I$ such that $\theta^* \notin D_T$. This means that the non-zero filters $L_{\theta^*}(z)$ and $M_{\theta^*}(z)$ are such that (6.1) holds. This defines, essentially, a linear, time invariant, noisefree relationship between y and u,

$$M_{\theta^*}(q^{-1})u(t) \sim L_{\theta^*}(q^{-1})y(t). \tag{6.3}$$

Therefore only if there is a feedback like (6.3) we may loose *SI*. This condition is the most general statement about System Identifiability, but as it stands it is fairly implicit. We shall therefore proceed to give an exact explicit condition for *SI*, which should cover most practical cases of interest.

Let the input signal, $u(t)$, to the process be given by

$$u(t) = F_i(q^{-1})y(t) + K_i(q^{-1})v(t) \qquad i = 1, \ldots, r \quad (6.4)$$

where F_i and K_i are linear rational functions of the delay operator q^{-1}. These filters are changed during the identification experiment between r (different) ones, so that each is used a non-zero proportion of the total time. The variable v is a vector of dimension n_v and it is an external disturbance to the input signal. Comparing with Fig. 1 we see that v can be considered as an additional extra input signal v_1 (measureable), as noise v_2 in the regulator, as set point changes in the regulator, v_3, or as a combination of these effects. To include the case where v is correlated with the system noise e, assume that it can be written

$$v(t) = \tilde{v}(t) + R(q^{-1})e(t) \quad (6.5)$$

where $\tilde{v}(t)$ and $e(s)$ are independent for all t and s and R is a causal filter. [More precisely, we shall assume that R is causal if there is a time delay in the feedback loop, and strictly causal ($R(0) = 0$) otherwise, see Theorem 6.1 below.] Before we give the result, we may remind the readers of the concept of persistently exciting signals, see, e.g., Åström–Bohlin [23] and Mayne [32]. We say that a signal $\tilde{v}(\cdot)$ is persistently exciting (*pe*) of order m if

$$\delta I < \frac{1}{N}\sum V_m(t)V_m(t)^T < \frac{1}{\delta}I \qquad N \geq N_0 \quad (6.6)$$

where

$$V_m(t) = \text{col}\,(\tilde{v}(t), \ldots, \tilde{v}(t-m+1)). \quad (6.7)$$

It is clear that if v is a stochastic process with full rank innovations, then it is *pe* of any finite order.

Theorem 6.1. Consider the system $\mathcal{S}$, (2.1) with input signal (6.4), $\mathcal{K}$, where $\tilde{v}$ is *pe* of any finite order. The model set may consist of arbitrary time invariant linear models as in (2.2). We assume there is a time delay either in the system (and model) or in the regulator, i.e. $G(0)F_i(0) = 0$. The identification method $\mathcal{I}$ is a prediction error identification method (2.5) applied in a direct fashion. Then $\mathcal{S}$ is Strictly System Identifiable, *SSI* ($\mathcal{I}$, $\mathcal{K}$) if and only if

$$\text{rank}\begin{bmatrix} K_1(z) \ldots K_r(z) & F_1(z) \ldots F_r(z) \\ 0 \quad \ldots \quad 0 & I \quad \ldots \quad I \end{bmatrix} = n_y + n_u \quad \text{a.e. } z \quad (6.8)$$

[The dimension of K_i is $n_u|n_v$, of F_i $n_u|n_y$, of 0 $n_y|n_v$ and of I $n_y|n_y$, so the dimension of the matrix in (6.8) is $(n_u + n_y)|r(n_v + n_y)$.] □

This theorem is an extension of the results in Ljung *et al.* [4]. The proof of the extended version is in Söderström *et al.* [33].

Remark. When the disturbance signal v is measurable, indirect identification may be applied, as described in Section 2. Then the transfer function(s) from v to y are determined (or, if $v = 0$, y is modelled as a time series) and knowing the regulator(s) the open loop dynamics is solved for. However, it can be shown, cf. [4] and [33], that the identifiability properties for this procedure are exactly the same as those for direct identification using (2.5), i.e. Theorem 6.1 is valid also for indirect identification. Moreover, in view of the discussion in Section 4.1, also joint input output identification will give the same result as direct identification in this case.

Obviously, a necessary condition for (6.8) to hold is that

$$r \geq (n_y + n_u)/(n_y + n_v). \quad (6.9)$$

On the other hand, if (6.9) holds then K_i and F_i can always be chosen so that (6.8) is satisfied.

Let us discuss some special cases of the result. First we see that if $n_v = n_u$ and $K_1(z)$ is non-singular, then (6.8) is satisfied for any r (including $r = 1$) regardless of the feedback filters F_i. Therefore, if the number of independent 'extra inputs' is equal to the number of inputs to the system, then we always have *SSI*, no matter what the feedback might be. This fact is by now also widely recognized, see below. If there are no extra inputs, $n_v = 0$, then (6.8) can be satisfied if and only if

$$r \geq 1 + n_u/n_y \quad (6.10)$$

by a proper choice of feedback laws. This means that by shifting between sufficiently many regulators we can also obtain *SSI* even if no extra input signal is allowed. This fact was pointed out in [4]. For example if $n_u = n_y$, then it is sufficient to use two regulators. These shall be chosen so that $\det(F_1(z) - F_2(z)) \not\equiv 0$, which is quite a mild condition. In certain applications an additional input would require a very good regulator in order to keep the output variance sufficiently low. In such cases it may be easier to find several different regulators that yield acceptable output variance, than to find one close to the optimum. Then the way of obtaining *SSI* by shifting regulators is of importance.

We may repeat that when (6.8) is satisfied, i.e. when the system is *SSI*, there is no additional difficulties in finding an appropriate model structure due to the feedback terms. The whole identification procedure can be performed as though the data was collected during open loop operation. In fact, the open loop situation is just a special case in the theorem ($F_i = 0$).

When (6.8) is not satisfied, then the system is not *SSI*. It may, however, be *SI* for certain model structures, but these can be characterized only in terms of fairly complex expressions involving the matrix in (6.8) and the structure of the true system $\mathcal{S}$. Neither can it be tested *a posteriori* from data only, whether the chosen $\mathcal{M}$ actually yielded *SI*, cf. Example 2.3.

An important case when *SSI* is impossible is when the regulator is time invariant, linear and noisefree and no extra input is added, i.e. $r = 1$, $n_v = 0$. We shall further investigate this case for $n_u = n_y = 1$ and $\mathcal{M}$ being difference equation models.

Let the system be given by

$$\mathcal{S}: y(t) + a_1 y(t-1) + \cdots + a_{n_a}y(t-n_a) = b_1 u(t-k-1) + \cdots + b_{n_b}u(t-k-n_b) + e(t) + c_1 e(t-1) + \cdots + c_{n_c}e(t-n_c) \quad (6.11)$$

$$k \geq 0,\ a_{n_a} \neq 0,\ b_1 \neq 0,\ b_{n_b} \neq 0,\ c_{n_c} \neq 0$$

and the regulator is

$$u(t) = -f_1 u(t-1) - \cdots - f_{n_f}u(t-n_f) + g_0 y(t) + \cdots + g_{n_g}y(t-n_g) \quad (6.12)$$

$$f_{n_f} \neq 0, \quad g_{n_g} \neq 0.$$

Define the polynomials $A(z)$, $B(z)$, $C(z)$, $F(z)$ and $G(z)$ by

$$A(z) = 1 + a_1 z + \cdots + a_{n_a}z^{n_a} \quad C(z),\ F(z) \text{ analogously}$$

$$B(z) = b_1 z + \cdots + b_{n_b}z^{n_b}$$

$$G(z) = g_0 + g_1 z + \cdots + g_{n_g}z^{n_g}.$$

Assume that $F(z)$ and $G(z)$ are relatively prime and that there is no common factor to all the polynomials $A(z)$, $B(z)$ and $C(z)$. Further assume that the polynomials $C(z)$ and $A(z)F(z) - z^k B(z)G(z)$ have exactly n_p common factors. Then we have the following result.

Theorem 6.2. (Söderström *et al.* [5]). Consider the system given by (6.11) with the feedback $\mathcal{K}$ (6.12). Let the class of models $\mathcal{M}$ also be given by (6.11), where the coefficients a_i, b_i and c_i are estimated using the identification method (2.5); $\mathcal{I}$. Then $\mathcal{S}$ is $SI(\mathcal{M}, \mathcal{I}, \mathcal{K})$ if and only if

$$\max(n_f - n_b, n_g + k - n_a) - n_p \geq 0. \quad (6.13)$$ □

A similar result is shown in Vorchik[34]; cf. also Box and MacGregor[35]. In [5] further results are given for the case when the time delay and the orders of the model do not coincide with those of the system. Results for the multivariable case are given in [87], [88].

Remark 1. In order to test identifiability of a system by (6.13) some condition invoking the true order of the system must be evaluated. Thus *a priori* information of the order of the system is a crucial point. Moreover, the order cannot be tested *a posteriori*, i.e. it is not possible to test after the experiments if the system really was identifiable as was illustrated by the Example 2.3.

Remark 2. The results can for example be applied to the special case when a minimum variance regulator, [36], is used. It can then easily be shown that (6.13) reduces to

$$k \geq n_c + 1.$$

6.2 *Influence on the accuracy.* From Section 6.1 it follows that good identifiability properties are obtained for a number of feedback situations. It is, however, sometimes claimed that even if identification of systems operating in closed loop is theoretically possible, the obtained estimates are in practice very poor, due to the effect of the feedback terms in the input signal. Normally, the reason for feedback control is to decrease the variance of the output. Then, naturally, also the information content of the measured data decreases and the estimates obtained have decreased accuracy. For this reason closed loop experiments may be inferior to open loop ones. However, this comparison is clearly unfair. Often the limiting variable for an identification experiment is the output variance. Due to security or production reasons this variance must be kept below a certain value. Therefore different experimental conditions should be compared for the same output variance.

Then the interesting questions to ask are:

—Are open loop experiments inherently better than closed loop ones giving the same output variance?

—Is there any systematic difference between various experimental conditions that all yield identifiability?

In Söderström *et al.*[6], [7], these two problems are studied by means of simulations and analysis of low order systems. The answers are quite clear for these examples: Open loop experiments are not necessarily better than closed loop ones that give the same output variance, and there is no systematic difference between different experimental conditions that yield identifiability. On the contrary, inclusion of feedback will increase the 'degrees of freedom' when choosing input signals.

We now proceed to optimize the experimental conditions in order to obtain as good accuracy of the parameter estimates as possible. We shall consider the following classes of experimental conditions.

$\mathscr{X}(O)$: Open loop experiments. The covariance function (or the spectrum) of the input signal is at the experiment designer's disposal.

$\mathscr{X}(S(r, n))$: Shifts between r linear regulators of order n. The parameters of the regulators, and the proportion of the total time each one is used are at disposal. ($r \geq 1$; $n \geq 0$).

$\mathscr{X}(E(n))$: A linear regulator of order n and an extra input signal. The parameters of the regulator as well as the covariance function of the extra input are at disposal.

To choose the optimal experimental conditions within the class $\mathscr{X}(O)$ is of course the well-known problem of optimal input design, which has been studied in a great number of papers, e.g. Levin[37], Levadi[38], Gagliardi[39], Aoki–Staley[40], Nahi–Napjus[31], Goodwin *et al.*[41], Keviczky–Bányász[42], Goodwin–Payne[43], Ljubojević[44] and Mehra[13]. In these papers the possibility to change the experimental conditions by, for example, introducing a feedback has not been considered.

The most general result concerning optimization of $\mathscr{X}$ for a general system seems to be given by Goodwin–Payne[87]. They consider minimization of det P for a general multivariable system under constraints on the input variance. If $G_\theta(q^{-1})$ and $H_\theta(q^{-1})$ in (2.2) have no common parameters then the optimal experimental conditions can always be found in the class of open loop experiments.

Unfortunately, it is technically involved to examine in more detail the optimization of $\mathscr{X}$ for a general system. We shall therefore analyse the problem only for a first order single-input single-output system.

$$\mathscr{S}_1\colon\ y(t) + ay(t-1) = bu(t-1) + e(t) + ce(t-1). \quad (6.14)$$

Assume that the model set $\mathscr{M}$ is chosen as

$$\mathscr{M}_1\colon\ y(t) + \hat{a}y(t-1) = \hat{b}u(t-1) + \epsilon(t) + \hat{c}\epsilon(t-1). \quad (6.15)$$

The identification method is chosen as a prediction error method applied in a direct fashion. As measures of the accuracy we shall use

$$V_1 = \det P \quad (6.16)$$

i.e. the determinant of the covariance matrix of the estimates and

$$V_2 = \operatorname{tr} QP \quad (6.17)$$

where Q is chosen so that V_2 corresponds to the expected increase in output variance if the process is controlled with a minimum variance strategy designed from the model, cf. Section 3.2, Example 3.3.

When no restriction is put on the variances of the input and the output, it would be favourable to have very large signals. This is of course for various reasons not possible in practice. Therefore the cases of constrained output variance,

$$\mathscr{C}_1\colon\ Ey^2(t) \leqq \lambda^2(1+\delta_1);\quad \delta_1 > 0 \quad \text{where} \quad \lambda^2 = Ee^2(t) \quad (6.18)$$

and constrained input variance,

$$\mathscr{C}_2\colon\ Eu^2(t) \leqq \lambda^2\delta_2/b^2;\quad \delta_2 > 0 \quad (6.19)$$

will be considered. Of these constraints it seems more realistic to put bounds on the variance of the output.

The optimal experimental conditions will, as expected be dependent on the choice of measure of accuracy and of the constraint. Also a change of the system or model structure can lead to a different result. In what follows we present part of the material from Söderström *et al.*[6], [7] and Söderström[60].

Example 6.1

Consider the system $\mathscr{S}_1$, (6.14) and the model structure $\mathscr{M}_1$, (6.15). Let $\{P(\mathscr{S}_1, \mathscr{M}_1, \{\mathscr{X}\})\}$ denote the set of covariance matrices that can be generated by choosing experimental conditions in the class $\{\mathscr{X}\}$. Then it can be shown that

$$\{P(\mathscr{S}_1, \mathscr{M}_1, \mathscr{X}(O))\} \subset \{P(\mathscr{S}_1, \mathscr{M}_1, \mathscr{X}(S(r, \cdot)))\} =$$
$$= \{P(\mathscr{S}_1, \mathscr{M}_1, \mathscr{X}(S(1, \cdot)))\} = \{P(\mathscr{S}_1, \mathscr{M}_1, \mathscr{X}(E(\cdot)))\}. \quad (6.20)$$

The set $\mathscr{X}(S(r, \cdot))$ consists of the experimental conditions with shift between r linear regulators of any order and similarly for $\mathscr{X}(E(\cdot))$. Note that in (6.20) the inclusion is trivial. The implication of the equality is interesting. No matter what measure of accuracy is used, as long as it is a function of the covariance matrix, shifts between r different regulators, as well as one regulator as well as a regulator with an extra input signal will all give the same optimum.

If the system and the model structure are restricted to $c = \hat{c} = 0$, still stronger results are available. Denote the system by $\mathscr{S}_2$ and the model structure by $\mathscr{M}_2$. Then,

$$\{P(\mathscr{S}_2, \mathscr{M}_2, \mathscr{X}^*)\} = \{P(\mathscr{S}_2, \mathscr{M}_2, \mathscr{X}(S(2, 0)))\} =$$
$$= \{P(\mathscr{S}_2, \mathscr{M}_2, \mathscr{X}(S(1, 1)))\} =$$
$$= \{P(\mathscr{S}_2, \mathscr{M}_2, \mathscr{X}(E(0)))\}. \quad (6.21)$$

In (6.21), $\mathscr{X}^*$ denotes the set of *all* possible experimental conditions with a causal feedback and such that the covariance matrix exists. The implication is that for *any* measure of accuracy it is always *sufficient* to consider either a shift between two proportional regulators or a regulator of

first order or a proportional regulator with an extra input signal. □

Example 6.2

Consider the system $\mathcal{S}_1$ (6.14). Assume that $V_1 = \det P$ is minimized subject to the constraint $\mathcal{C}_1$. Then it can be shown that the optimal experimental condition in the class $\mathcal{X}(E(\cdot))$ is obtained as

$$u(t) = \frac{a-c}{b} y(t) + v(t) \tag{6.22}$$

where $v(t)$ is white noise with variance $\delta_1(1-c^2)\lambda^2$. It is consequently sufficient to take $n = 0$.

In the class $\mathcal{X}(S(r, \cdot))$ the optimal experimental condition will be

$$u(t) = \frac{a}{b}\left(1 - \frac{c}{1+\delta_1} \pm \frac{\sqrt{[\delta_1(1-c^2+\delta_1)]}}{1+\delta_1}\right) y(t) \tag{6.23}$$

where the + sign and − sign are each used 50% of the time. Hence r can be taken as 2 and n as 0. The two proportional regulators in (6.23) will both meet the constraint $\mathcal{C}_1$ precisely. According to (6.20) both (6.22) and (6.23) will give the same optimal value of V_1. It can also be shown that *no* open loop input can give this optimum, except for the degenerated case $a = c$. Moreover, for the considered case the same result applies also to the criterion V_2, (6.17).

For the system $\mathcal{S}_2$ and the model structure $\mathcal{M}_2$ the corresponding results are similar and the only essential difference is that optimum can be obtained in $\mathcal{X}(O)$ if and only if $\delta_1 \geqq |a|/(1-|a|)$. □

Remark. In [89] the results of Example 6.2 are somewhat generalized. It is shown that a minimum variance regulator and a white noise input, cf. (6.22), will give the optimum of $\det P$ under the constraint $\mathcal{C}_1$ also for the system

$$y(t) + a_1 y(t-1) + \cdots + a_n y(t-n) = bu(t-1) + e(t).$$

Although we consider it to be more natural to require that the variance of the output is bounded than that the variance of the input is constrained, it is instructive to see how this change of constraint will effect the result.

Example 6.3. Consider the system $\mathcal{S}_2$ and the model structure $\mathcal{M}_2$. Assume that the criterion V_1 is to be minimized under the constraint $\mathcal{C}_2$. Due to (6.21) the overall optimum can be obtained in $\mathcal{X}(S(2,0))$, in $\mathcal{X}(S(1,1))$ and in $\mathcal{X}(E(0))$. Now it can be shown that the optimum can also be obtained in $\mathcal{X}(O)$, cf. [87] for general results on the constrained input variance case, with the choice

$$u(t) = \frac{1}{1+aq^{-1}} w(t)$$

where $w(t)$ is white noise with variance $\delta_2(1-a^2)/b^2\lambda^2$.

In the case $\mathcal{X}(S(2,0))$ the optimum is obtained with shifts between the two proportional regulators, each used 50% of the total time,

$$u(t) = \frac{-a\delta_2 \pm \sqrt{[\delta_2(1+\delta_2-a^2)]}}{b(1+\delta_2)} y(t). \tag{6.24}$$

If instead optimization of the criterion V_2 is considered, the result will be quite different. Open loop operation is only sometimes possible for achieving an optimum. The precise condition is an involved relation between a and δ_2. In $\mathcal{X}(S(2,0))$ still the regulator (6.24) is to be used, but the + and − signs must be used in different proportions of the total time. □

The analysis has several important implications. In the first place it shows that feedback terms may have to be included in the input signal in order to obtain optimal accuracy. The optimal input problem, as usually discussed, therefore can be said not to be well posed. Inclusion of feedback terms is by no means any problem in practice, and therefore the optimal input should be sought in the classes $\mathcal{X}(S(\cdot,\cdot))$ or $\mathcal{X}(E(\cdot))$ rather than in $\mathcal{X}(O)$.

One might discuss the generality of the results in Example 6.2. It is difficult to carry out the analysis in the general case, but it seems plausible that it is always favourable to control the process with a minimum variance controller and add an extra input so that the constraint on the output variance is met. Now, to know the minimum variance controller is essentially equivalent to know the system itself and then there is no need for identification. This is the usual dilemma of optimal input synthesis: The system must be known in order to design the optimal input signal. However, the following rule-of-thumb can be given: Decrease the output variance as much as possible using a feedback regulator. This makes use of the available *a priori* information about the system in a sensible manner. Then add an extra perturbation signal, e.g. white noise, so that the output variance reaches its allowed value. Alternatively, change the regulator parameters in two different ways so that the bound on the variance of the output is met. Then shift between these two regulator settings. One way to obtain a suitable regulator in the first place if the process dynamics are unknown could be to use a self-tuning regulator, Åström and Wittenmark [45].

6.3 *Survey of other contributions.* A number of papers dealing with the problem of identifying a process operating in closed loop are now available. So far most of them discuss the problem from the identifiability point of view. In the following the references will be briefly reviewed. In a short review like this it is not possible to give full details. Notions like identifiability etc will therefore be used without giving the definitions that were used in each particular paper. Nevertheless it is most often intuitively clear what is meant. In Table 1 an attempt is made to systemize the contributions according to the cases discussed and the methods and model structures used. The papers are reviewed in chronological order.

Fisher [14] discusses identifiability of continuous, deterministic, linear state space systems with no extra perturbation and with a time invariant, noise free feedback. A least squares approach to the parameter estimation is taken. Necessary and sufficient conditions for identifiability are in this case that the control law is non-linear in the states, and that the system is completely controllable.

The problem of identification of closed loop systems has been treated by Akaike [9], [17]. There are also several applications using his ideas, e.g. Otomo *et al.* [46] and Itoh *et al.* [47]. Akaike points out that cross spectral analysis requires that the input is measured without noise and that the noise and the input are independent. This last condition is violated for closed loop systems. If there is an extra perturbation, i.e. $v_1 \neq 0$, cross spectral analysis can, however, be used in a special way. Akaike treats particularly the case when there is an unmeasurable disturbance in the feedback loop. Then ordinary cross spectral analysis fails. Instead the problem is solved by introducing a causal time domain model. Direct identification is used with an impusle response model with a finite number of parameters.

Priestley [19] treats the problem of estimating the process transfer function from data consisting of records of the input and the output from a closed loop system with additive disturbances. No known perturbation is available and the feedback is linear. Direct identification is used. The main objective is to find conditions under which the least squares and the weighted least squares estimates of the system transfer function are identical, or reasonably close to each other. Analytical expressions for this equivalence and for the bias of these estimates are given.

Schwalm [70] determines the frequency response function of a linear feedback system with a random (measurable) input signal by means of correlation analysis. However, in order to estimate this function it is assumed that open loop measurements also can be performed.

In the survey paper on identification by Åström and Ekyhoff [10] identification of closed loop systems is only discussed briefly. An example is given for the case of noise free, linear, time invariant regulator without extra perturbation. In this case a noncausal method will give the inverse of the regulator as the model. However, a method using a causal model may or may not give the correct model, cf. Example 2.1 and 2.2.

TABLE 1. CLASSIFICATION OF SOME PAPERS DEALING WITH THEORETICAL ASPECTS OF IDENTIFICATION OF CLOSED LOOP SYSTEMS. PARENTHESES INDICATE THAT ONLY A SHORT DISCUSSION IS GIVEN

	Feedback cases					Methods					Systems		Model structures			
Author(s)	EI	NR	LTI	NL	LTV	D	ID	JIO	PE	COR	SI SO	MI MO	SSP	DE	IR	FR
Akaike [9], [17]	×	×				×	(×)	(×)	×	×	×	×			×	×
Bohlin [12], [48]	×		×			×	(×)		×	(×)	×			×		
Box–MacGregor [11], [18]	×	×	×			×	×		×	×	×			×	×	
Box–MacGregor [35]	×		×			×			×		×			×		
Caines–Chan [1], [2]		×						×	×		×	×		×		
Caines–Wall [50]		×						×	×		×	×		×		
Chan [93]		×						×	×		×	×		(×)		
Defalque *et al.* [68]	×								×	×	×		×		×	
Eykhoff [52]	×		(×)			×	×		×		×			×	(×)	
Fisher [14]				×		×			×		×		×			
Froisy *et al.* [69]			×			×			×		×			×		
Gevers [92]		×						×			×	×		(×)		
Glover [15]	(×)		×				×		×		×	×	×			
Goodwin *et al.* [53]	×						×		×		×			×		
Goodwin *et al.* [88]	×	×	×			×		×	×		×	×		×	×	
Graupe [62]	×	×				×			×		×			×		
Gustavsson *et al.* [3]	×	×	×	(×)	×	×	×		×	(×)	×	×		×		
Kurz–Isermann [59]	×	×	×		(×)	×	×		×	×	×			×		
Leonhard [51]	×	×	×			(×)	×		×		×			×		
Lindberger [54], [55]		×	(×)				×		×		×			×		
Ljung *et al.* [4]					×	×	×		×		×	×	(×)	(×)	(×)	(×)
Panich–Trachevskii [56]			×			×			×		×			×		
Phadke–Wu [58]		×						×	×		×	×		×		
Priestley [19]		×	×			×			×	×	×				×	×
Rödder [20], [21]	×	×	×				×			×	×					×
Saridis–Lobbia [16]	×					×			×		×		×			
Schultze [49]	×	×				×			×	(×)	×			×		
Schwalm [70]	×						×			×	×					×
Söderström *et al.* [5]			×			×			×		×			×		
Söderström *et al.* [6]	×	×	×		×	×	×		×		×	×	(×)	×	(×)	(×)
Thöm–Krebs [22]		×				×	×		×	×	×			×		×
Wellstead–Edmunds [63]		×	×			×			×		×			×		
Vorchik [34]	×	×	×			×			×		×			×		
Vorchik *et al.* [57]		×				×			×		×			×		

Feedback cases: EI: Extra input, the signal v_1 in Fig. 1 is non-zero; NR: noise in the regulator, the signal v_2 in Fig. 1 is non-zero; LTI: linear, time-invariant, noisefree regulator without extra input; NL: non-linear regulator; LTV: linear time-varying regulator. *Identification method* (cf. Section 2.1): D: Direct identification; ID: indirect identificatioh; JIO: joint input-output identification; PE: prediction error identification method (e.g. ML or LS); COR: correlation or spectral analysis. *System*: SISO: Single input single output; MIMO: multiple input multiple output. *Model structures*: SSP: State space model; DE: difference equation model; IR: impulse response model; FR: frequency response model.

A general discussion is given by Bohlin[12]. This paper is concerned with the basic limitations of identification and the practical implications of the mathematical assumptions involved. Thus also identification of processes operating in closed loop during the experiments is treated. Bohlin remarks that there are very simple but realistic cases when identification is impossible under closed loop conditions. However, a closed loop during the experiments does not necessarily prevent identification of the open loop characteristics. In the case of noise free, linear feedback without any extra perturbation the process is not identifiable without *a priori* knowledge of the structure. A chosen structure cannot be validated by the data, cf. Example 2.3. An application to a paper machine is given.

In Bohlin[48] the ambiguity of the maximum likelihood method is treated. It is demonstrated that if the regulator is noise free, linear, time invariant and if no extra perturbation is injected only the noise transfer function for the closed loop system is identifiable. When the structure is known, identifiability of the open loop transfer function is secured if there is a one to one correspondence between the parameters of the open loop transfer function and the closed loop noise transfer function. No explicit conditions for the identifiability of the open loop dynamics are given. An application to paper machine data is used to illustrate the results.

Schultze[49] treats the case with an extra perturbation v_1 or with an unknown disturbance v_2. The choice of model structure and identification criterion is also discussed. It is demonstrated that this choice is of crucial importance. For the closed loop case the difference equation model for the process must include possibilities to handle correlated disturbances. Direct identification is proposed. An application to a ball mill is presented.

Box and MacGregor[11], [18] first show that correlation methods fail for closed loop systems. If an extra input signal is available however identification is possible. They also discuss what can be done in the case of no extra input by modelling the process as a pure time series using the output, i.e. indirect identification. They propose a test for feedback by testing the cross covariance between the input and output. Applications to polymer viscosity and paper machine data are given.

Caines and Wall[50] discuss parameter estimation of closed loop systems and particularly the case with an unknown disturbance in the feedback loop. They state that direct identification using the loss function for the open loop maximum likelihood estimation will not give the maximum likelihood estimates for the closed loop case. Instead it is proposed to estimate the parameters in the model consisting of the inputs and outputs expressed as time series of the noise sources, i.e. joint input output identification. This idea has been pursued by Caines and Chan, see below.

Leonhard[51] discusses the identification of systems operating in closed loop. Indirect identification with the least squares method is used. It is stated that the open loop transfer function can be obtained if an extra perturbation v_1 is injected or when there is a disturbance v_2 acting in the feedback loop. An example using direct identification is also given.

Lindberger[54], [55] has discussed problems related to the identification of closed loop systems. In principle the case with unmeasurable disturbances in the feedback loop is treated. Lindberger uses indirect identification according to our notion. He also proposes a strategy in order to try to achieve identifiability by successively increasing the complexity of the regulator and performing one experiment with each regulator until a reasonable model is obtained.

Glover[15] is mainly concerned with parameter identifiability of continuous and discrete time state space systems. A recommended way to model a system with feedback is to write down the state space equations for the open loop system and then modify these equations with the feedback law. The closed loop system is then expressed in terms of the open loop parameters and the parameters of the feedback law. The identifiability questions for the unknown parameters of this closed loop system can then be answered as for the open loop case.

Goodwin, Payne and Murdoch[53] are mainly concerned with the synthesis of optimal test signals for closed loop identification. A variant of the case with an extra perturbation is treated. According to our notion indirect identification is used.

Panich and Trachevskii[56] derive identifiability conditions in the case of a linear noisefree feedback without any extra input assuming special structures of the system and of the regulator making least squares identification applicable. Necessary conditions turn out to be that the lag of the system is large enough or that the controller is complex enough. Simulations illustrate what happens for different number of delays in the system and for different controllers.

Vorchik, *et al.*[57] consider the identification of a closed loop system with disturbances in the feedback loop which may be correlated with the system disturbances. The disturbances are assumed to be uncorrelated in time. Conditions on the signals and the regulator are derived, under which the parameters of the system and/or the regulator converge to the true values and under what conditions the estimates are efficient. These results are extended in Vorchik[34] to cases where the structure of the disturbances is more complex. Identifiability conditions are also given for the case when there is no disturbance source in the feedback loop.

Rödder[20], [21] gives conditions on which signals that must exist and which signals that have to be measured to be able to determine the open loop characteristics from closed loop experiments by correlation (spectral) analysis. In general cases the regulator has to be known. Also estimates of the obtained accuracies are given.

Eykhoff[52] discusses briefly identification of closed loop systems in connection with least squares estimation. The case with noise free feedback and with a perturbation signal is treated. Indirect and direct identification are proposed as the two existing possibilities. It is mentioned that difficulties can arise in indirect identification because it is not always trivial to solve for the open loop dynamics from the closed loop transfer function.

Phadke and Wu[58] present a procedure for the identification of a multivariable system described by a vector difference equation model which is similar to the one proposed by Caines and Chan[1]. The procedure consists of two steps. First a multivariate time series model is fitted to the input and output series. The open loop characteristics can then be determined from this model. The conditions are that there is a disturbance source in the feedback loop and that the plant (or feedback) has at least one lag. The procedure is applied to the identification of a multivariable model of a blast furnace.

Froisy *et al.*[69] use least squares technique to identify closed loop systems. As a first step correlation functions are estimated. They are then used in a second step for the estimation of a parametric model. The design of controllers to give suitable experimental conditions for this identification technique is also discussed.

Thöm and Krebs[22] compare correlation analysis and parameter estimation when applied to the identification of closed loop systems. They discuss the possibilities to use correlation analysis and conclude that e.g. generalized least squares and maximum likelihood methods are preferable.

Kurz and Isermann[59] consider several different closed loop configurations and discuss under what conditions the open loop characteristics are identifiable. A two-stage (on-line) identification procedure is used. In the first step correlation analysis is performed. The obtained estimates of the correlation functions are then used in a least squares parameter estimation. The different cases are illustrated by simulations trying to show the achievable accuracy in the different cases.

Box and MacGregor[35] study the effects of optimality and suboptimality of the regulator on the estimation of characteristics of a system operating in closed loop. They also consider the effects of an additional input signal and of lags in the system. In particular, they consider two problems, the estimation of parameters in the dynamic and stochastic parts of the system and the estimation of only those functions of

these parameters which occur in the control equation. Identifiability conditions are given showing that the conditions for the latter case are much weaker.

Caines and Chan[1] and especially[2] give a very complete treatment of the problem with feedback in input signal. They present a definition of feedback between an ordered pair of multivariable processes. It provides statistical criteria for the detection of feedback. The proposed identification procedure starts with a multivariable time series, modelling the input and output. It is assumed that there is a disturbance in the feedback loop. From the obtained model the system and regulator characteristics can be derived. The papers include an application to economic model building. An application to power systems identification is described in [2] and further developed in Caines and Sinha[61].

Graupe[62] considers the identification of closed loop feedback systems of different configurations and discusses consistency of the obtained parameter estimates.

Wellstead and Edmunds[63] consider the identifiability problem in the case when the system has uncorrelated disturbances. They also discuss the case when there is no lag either in the system or in the regulator. It is proposed that the identification in this case can be performed using an instrumental variable method.

Goodwin and Payne[87] and Goodwin *et al.*[88] discuss various aspects of identification of closed loop systems. An interesting result presented in [87] is that the optimal experimental condition can always be found in the class of open loop experiments if the criterion is the determinant of the covariance matrix and if there is a constraint on the input variance. In [88] sufficient conditions are established for identifiability of a multiple input multiple output system with a noisefree feedback.

Ng *et al.*[89] show that for a special class of systems the optimal experimental condition for the case of output power constraint is achieved by a minimum variance controller with a white noise external input v_1.

Defalque *et al.*[68] consider identification of a single input single output feedback controlled linear system perturbed by coloured noise. They use a two stage algorithm which does not require any assumptions on the structure of the regulator. First the input–output dynamics are identified using a stochastic approximation algorithm which requires a white noise input v_1. In the next step the model of the disturbances is obtained using stochastic realization theory.

Gevers[92] reviews some identifiability results for closed loop identification and analyses particularly the joint input output identification method. A set of sufficient conditions are given under which the open loop transfer function and the noise dynamics can be uniquely and correctly determined from the joint process model. Similar results have been published by Chan[93] and Goodwin *et al.*[88]. Gevers also notes that the joint input output identification method in some cases is preferable to a direct approach because the model can be identified with covariance factorization which does not require an *a priori* parameterization.

So far the papers reviewed have mainly been concerned with the identification of the open loop dynamics from closed loop experiments with constant regulators. Identifiability problems for closed loop systems appear also naturally in many adaptive control situations. For such problems the feedback is generally timevarying in a very subtle way. Some papers have discussed such problems, e.g. Turtle and Phillipson[64], Saridis and Lobbia[16], Lobbia and Saridis[65], Balakrishnan[66], Ljung and Wittenmark[67].

7. *Applications*

In this section two applications of identification of processes operating in closed loop during the experiments will be described. Other applications found in the literature will be surveyed in the second part of this section.

7.1 *Applications to ship dynamics and to a laboratory process.*

Example 7.1. This application is concerned with the identification of ship dynamics. A model of the ship dynamics is needed for example when designing autopilots or for simulation. In this case it is valuable to be able to use data from closed loop experiments for the identification, since the ship then can operate under fairly normal conditions even while the experiments are carried out.

Two experiments with a fully loaded 255000 tdw supertanker, *T/T Sea Swift*, are compared. The ship is 329 m long and has a maximum speed of 16 knots. The measurements were made in the Indian Ocean. The weather conditions, the trim of the ship and the water depth were about the same in the two experiments. The dynamics from requested rudder angle to the ship's heading angle was determined. The first experiment was performed in open loop, the second one in closed loop with a proportional regulator and with an additive rudder disturbance in order to secure the identifiability. The open loop experiment lasted 78 min, the closed loop one 59 min. The sampling interval was chosen as 10 seconds. The variance of the input signal was approximatively 4 times larger in the closed loop experiment. On the other hand the variance of the heading angle for the closed loop experiment was one fourth of the corresponding variance for the open loop

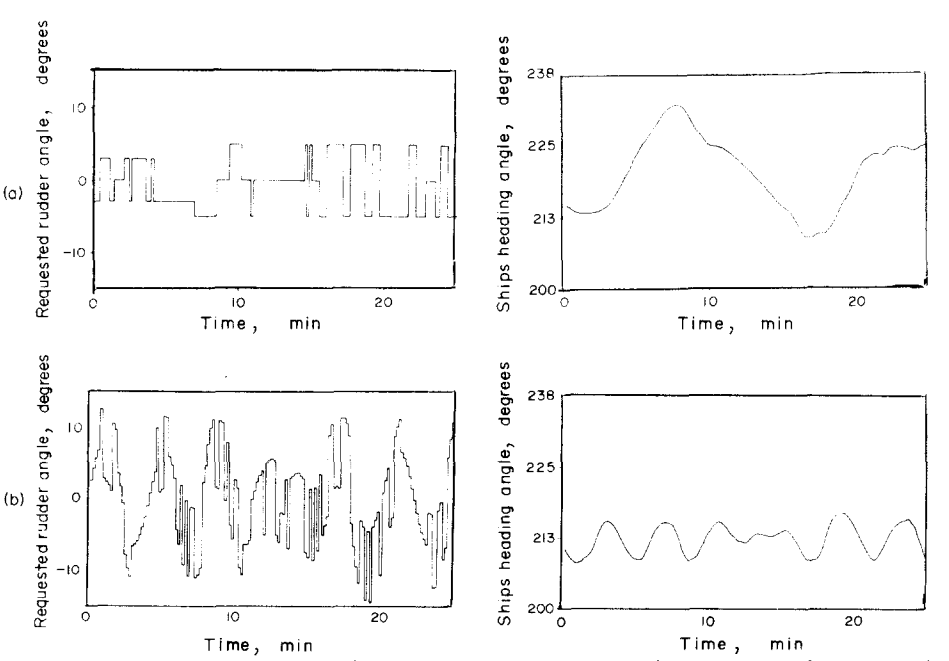


FIG. 3(a). Requested rudder angle and heading angle for the open loop experiment. (b) Same for the closed loop experiment.

experiment. In Fig. 3 the inputs and outputs are shown for the two experiments. The experiments are described in more detail in Källström[71].

The dynamics from the requested rudder angle to the differenced heading angle was determined in a model form like (6.11) using the maximum likelihood method. (The reason for using the differenced heading angle as the output instead of the angle itself, is that, due to the physical knowledge of the process, it is known that there is a pure integrator in the process, when no wind is present, viz. the heading angle is the integrated angular velocity of the ship[72].)

For both experiments statistical tests indicated that a third order model was appropriate. Also cross correlation functions between the residuals and the input indicated a third order model. The parameters of the models are given in Table 2 together with the estimated accuracies of the parameters.

The models do not differ very much. The zeros and the poles of the corresponding continuous models are given in Table 3.

There is one unstable mode in the model. This is not unexpected since it was known in advance that the ship might be unstable under certain loading conditions. Furthermore a pair of complex poles occur. These modes may be due to nonlinear effects in the ship dynamics. Another explanation may possibly be the dynamics of the rudder servo.

In conclusion we may say that the two experiments are quite comparable both regarding results and accuracies. However during the closed loop experiments the variance of the output was considerably less. The disturbances in the course of the ship would be quite acceptable for longer experiments, when the control loop is closed.

Example 7.2. In this case a laboratory process, a bar with a rolling ball is identified, see Fig. 4. The angle of the bar and the position of the ball can be measured. The control variable is the voltage of the motor driving the bar around.

In this case we were interested in the dynamics from the angle φ to the position x. It is clear from basic physical laws

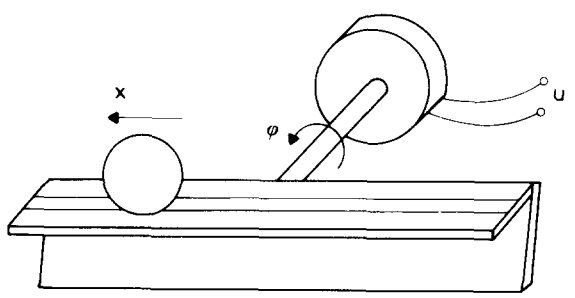


FIG. 4. The process: A bar, which can be rotated with a motor around a horizontal axis, applied in its midpoint. A metal ball rolls freely along the bar.

TABLE 2. THE ESTIMATED PARAMETERS OF THE THIRD ORDER DIFFERENCE EQUATION MODELS OF THE SHIP DYNAMICS. THE b-PARAMETERS ARE SCALED WITH A SCALE FACTOR 100

	Open loop experiment	Closed loop experiment
$\hat{a}_1$	-2.031 ± 0.051	-2.012 ± 0.073
$\hat{a}_2$	1.322 ± 0.074	1.337 ± 0.088
$\hat{a}_3$	-0.297 ± 0.037	-0.330 ± 0.034
$\hat{b}_1$	-0.649 ± 0.110	-0.526 ± 0.132
$\hat{b}_2$	-1.381 ± 0.230	-2.146 ± 0.248
$\hat{b}_3$	1.355 ± 0.196	1.702 ± 0.273
$\hat{c}_1$	-1.426 ± 0.058	-1.284 ± 0.088
$\hat{c}_2$	0.603 ± 0.041	0.484 ± 0.057
$\hat{\lambda}$	0.078	0.095

TABLE 3. POLES AND ZEROS OF THE CONTINUOUS MODELS OF THE SHIP DYNAMICS

	Open loop experiment	Closed loop experiment
Zeros	-0.031	-0.038
	0.31	0.25
Poles	0.0020	0.0015
	$-0.062 \pm 0.036i$	$-0.056 \pm 0.051i$

that this dynamics is a double integrator: $G(s) = 1/s^2$. Obviously, the identification experiment cannot be performed in open loop. With no control, the ball will very quickly roll off the bar. A simple *PD*-controller feeding back x to the voltage of the motor was used. A disturbance was added to the setpoint of this controller, so that the ball was made to roll from one end of the bar to the other. Inputs (φ) and outputs (x) are shown in Fig. 5.

The sampling interval was 0.04s and the experiment lasted 20 sec. The parameters of a second order difference model (6.11) were determined using the maximum likelihood method. The model was converted to continuous time, and an amplitude Bode plot was drawn, Fig. 6. It is seen that in a wide frequency range the model gives a good description of the system $1/s^2$. The 'bad' behaviour for very low frequencies is explained by the fact that very little input energy is put in at these frequencies, cf. Fig. 5. This reveals a dilemma with closed loop experiments: in order to keep the ball on the bar, the regulator must be very good at damping slow modes. Therefore we will get only little information about these, and the model will be bad for low frequencies. The experiments are further described in Wieslander[73].

7.2 *Survey of applications.* Some papers dealing with application of identification of closed loop systems are classified in Table 4. As can be seen from the table

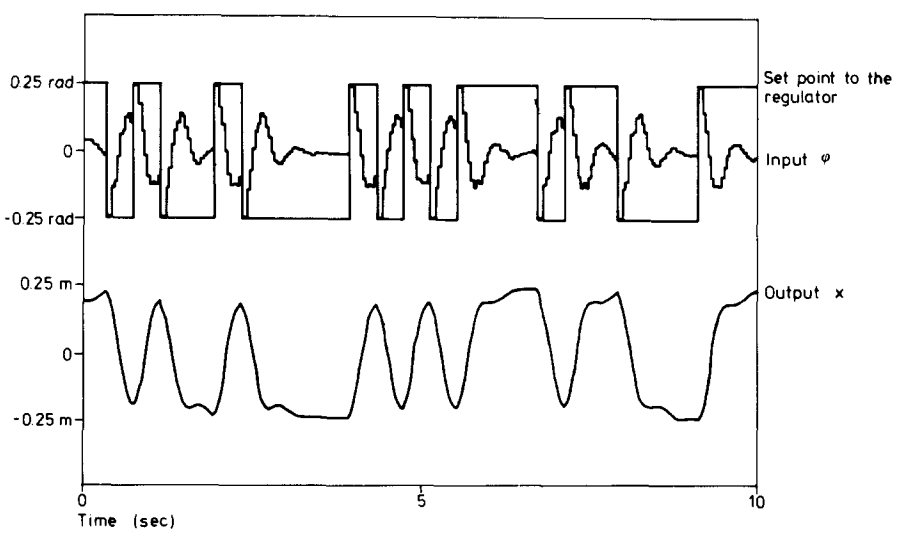


FIG. 5. Inputs (φ) and outputs (x) for the identification experiment.

TABLE 4. CLASSIFICATION OF SOME PAPERS DEALING WITH APPLICATION OF IDENTIFICATION OF CLOSED LOOP SYSTEMS (SEE TABLE 1 FOR EXPLANATION)

Author(s)	Application	Feedback cases					Methods					Systems		Model structures			
		EI	NR	LTI	NL	LTV	D	ID	JIO	PE	COR	SI SO	MI MO	SSP	DE	IR	FR
Bacon–Howe [80]	Evaporator	×	×					×			×	×			×		
Bohlin [12], [48]	Paper machine	×					×			×		×			×		
Box–MacGregor [11], [18]	Paper machine Polymer viscosity	×					×	×		×	×	×			×		
Breddermann *et al.* [90]	Human operator	×					×	×		×	×	×			×		×
Caines–Chan [1], [2]	Econometrics		×						×	×			×		×		
Caines–Sinha [61]	Power generator		×						×	×			×		×		
Dablemont–Gevers [91]	Glass furnace		×						×				×		×		
Hong–MacGregor [79]	Steam-jacketed stirred tank process	×					×			×		×			×		
Itoh *et al.* [47]	Cement kiln		×				×			×		×					×
Lindahl–Ljung [83]	Power generator		×						×	×			×	×			
Okomo *et al.* [46]	Cement kiln		×						×	×	(×)	×			×		
Phadke–Wu [58]	Blast furnace		×						×	×			×		×		
Phadke–Wu [74]	Papermaking process		×						×	×		×			×		
Rake [76], [77]	Blast furnace	×						×			×	×					×
Schultze [49]	Ball mill	×	×				×			×		×				×	×
Shinners [81]	Human operator	×						×			×	×			×		
Sutherland-Walker [82]	Power system	×						×			×	×					×
Wall [78]	Econometrics		×						×	×			×		×		
Tee-Wu [75]	Papermaking process		×				×				×	×					×

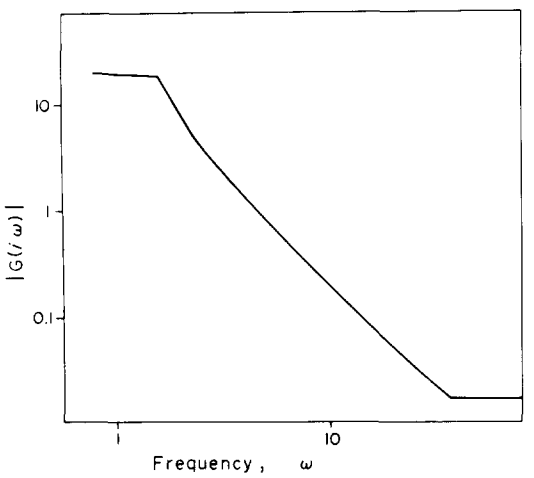


FIG. 6. Amplitude Bode-plot of the frequency response of the obtained model, converted to continuous time.

applications can be found to blast furnaces, glass furnaces, cement kilns and paper machines, to some chemical processes, to economic systems and to the identification of the human dynamics. These applications should be regarded as typical since such processes are either difficult or expensive to operate without a controller or it is impossible to break the feedback loop.

The systems have been both of single input single output type and multivariable, in most of the cases difference equation models are used. This also implies that causal parametric methods have been used with only some exceptions. Direct as well as indirect identification techniques have been applied. It is also striking to see that in almost every case an extra input signal has been introduced or it has been assumed that there is a disturbance in the feedback loop. This is of course a consequence of the fact that for these cases identifiability is generally secured.

The papers that have not already been reviewed in Section 6.3 will now be very briefly described.

Rake[76], [77] determines the dynamics of a part of a blast furnace by estimating the frequency response functions using normal operating data. The furnace is operating in closed loop because of control actions of the personnel. An attempt is made to eliminate the influence of this feedback.

Bacon and Howe[80] estimate parametric models for an industrial evaporator using routine unperturbed operating data. Their main objectives were to evaluate the existing control system and to modify it.

Tee and Wu[75] and Phadke and Wu[74] identify the process dynamics of a paper machine. The data was collected with an operator as feedback element. In [74] a joint input output identification technique is used.

Lindahl and Ljung[83] identify the dynamics of a power generator using normal operating data. Different identification methods are used. Results from spectral analysis and maximum likelihood identification are compared. Also a joint input output identification method is proposed and applied to the data.

Shinners[81] determines the dynamics of the human operator from data obtained under closed loop operation. Indirect identification is used, and there is an external, measurable, random input signal which is utilized.

Wall[78] uses joint input output identification technique to detect feedback and to estimate a model for the money–national income part of the U.K. economy.

Hong and MacGregor[79] discuss identification and direct digital control of a steam-jacketed continuous stirred tank process. The data are collected under closed loop operation and a direct identification technique is applied. The paper also includes a short discussion on design of experiments for closed loop systems.

Sutherland and Walker[82] use pseudo-random binary sequence perturbations to identify the dynamics of a boiler turbine unit. The experiments had to be performed with the automatic control loops closed. An indirect identification method was used which utilized the fact that a measurable external input signal was applied. Correlation functions and frequency responses were estimated.

Breddermann *et al.*[90] is another example of identification of the dynamics of a human operator. Several different identification methods, such as correlation technique, the generalized least squares method and the maximum likelihood method, are compared.

Dablemont and Gevers[91] identify the dynamics of an industrial flat glass furnace melter using a joint input output identification method. The resulting model is used on a process control computer for automatic control of the melter.

8. *Conclusions*

Several approaches to identification of closed loop systems have been taken. We have mainly studied the use of direct identification with a prediction error method applied to general linear models. This is a 'good' method in the sense that if it fails, then no other method will succeed, using the same data. A further advantage is that no special procedures or programs have to be devised that take the specific feedback situation into account. It has been shown that identification is possible in a variety of feedback situations; e.g. when an extra input signal is applied, when the set point to the regulator is varied, when there is noise in the regulator, or when the regulator is time-varying or non-linear. In these cases the system is "Strongly System Identifiable" (SSI) which means that the identifiability properties are as good as for open loop experiments, and the fact that the data have been collected during closed loop operation can be disregarded. Even for a linear, time-invariant and noise-free regulator the system may be identifiable, although it is not SSI. Some conditions in a special case are given in Theorem 6.2.

Furthermore, our analysis has indicated that identification of closed loop systems is not only theoretically possible; neither is it necessarily inferior to open loop experiments as regards accuracy. In fact, for experiments with constrained output variance, it has been shown that input signals containing feedback may yield strictly better accuracy than can be obtained with open loop experiments. This fact obviously has some important implications for optimal input design.

Several practical applications have also demonstrated that closed loop identification experiments, which often are a necessity, in fact do give acceptable results both from the identifiability and accuracy points of view.

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